3 Yr. Degree/4 Yr. Honours 4th Semester Examination, 2025 (CCFUP)

Subject : Computer Science

Course: COMP4012 (MAJOR)

Time: 2 Hours **Full Marks: 40**

The figures in the right hand margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

1. Answer *any five* questions: 2×5=10

(a) What is Just-in-Time compiler in Java?

(b) Why main ( ) method in Java is public and static?

(c) What are command line arguments in Java?

(d) What is wrapper class in Java?

(e) What are the default values assigned to variables and instances in Java?

(f) Differentiate between System.out and System.in

(g) What is the main difference between String and String Buffer classes.

(h) What do you mean by Event Handling Mechanism in Java?

2. Answer *any two* questions: 5×2=10

(a) What is package? Create one user defined package and use it in a simple Java program. 1+4

(b) Explain a few object oriented features of Java. Why Java is called fully object oriented language? 4+1

(c) How can you create an array in Java? Write a program in Java to sort a set of random numbers (use util package) 1+4

(d) What is exception? How does Java handle exception? 1+4

3. Answer *any two* questions: 10×2=20

(a) What is applet? Write a program in Java that creates a banner and then creates a thread to scroll the message in the banner from left to right accross the applet's window. 1+9

(b) What are the different ways to implement multithreading in Java? Write a program to demonstrate priorities among multiple threads. 4+6

(c) What is method overloading? Explain it with an example. Explain the purpose of final keyword in Java. 2+5+3

(d) What is Garbage collection? What is the purpose of garbage collection? What are the interfaces in Java? How do they differ from class? 2+3+3+2